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NCC education

Qualification: L5DC-BM

Unit Name: Information Systems and Organisations

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Assignment Marking Sheet

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Centre Name	Talent Nurtures Educational Institute
Candidate Name	Mhd Laith Alkhateeb
NCC Education Candidate ID No.	213142
Examination Date	15/8/2025
Examination Unit Title	Information Systems and Organisations

Questions answered by the candidate:

Please insert the question numbers you have answered in the table below.

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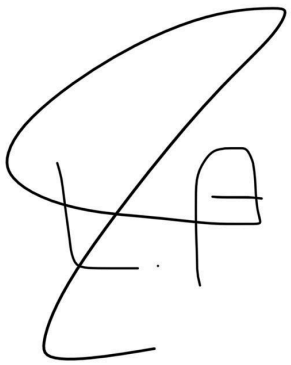
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Candidate Name:	Mhd Laith Alkhateeb
Candidate ID Number:	213142
Qualification:	L5DC-BM
Unit:	Information Systems and Organisations
Centre:	Talent Nurtures Educational Institute
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Information Systems and Organisations

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Overview

One of the biggest suppliers of cloud services and the biggest online retailer in the world is Amazon (Amazon.com). By 2025, it is known as a giant in cloud computing and e-commerce.

Amazon, which has its headquarters in Seattle, has fulfillment centers, data centers, software development facilities, and individual websites all over the world. In 1994, Jeff Bezos started the corporation, and until 2021, he served as its president and CEO. He is still one of Amazon's biggest individual shareholders and serves as executive chair as of 2025.

Amazon began as an online book seller but has now evolved into a web-based company that offers e-commerce, cloud computing, digital streaming, and artificial intelligence services.

The platform is active in over 20 countries across the globe. Each year, hundreds of millions of customers are served by these bustling Amazon marketplaces, which have an enormous inventory and product selection. They enable customers to purchase a wide range of items, such as apparel, cosmetics, fine dining, jewelry, books, movies, electronics, pet supplies, furniture, toys, garden supplies, and household items.

Amazon became well-known as a cloud service provider in more recent years. In order to offer businesses cloud computing services including computation, storage, databases, and networking, the company founded Amazon Web Services (AWS) in 2006. With these products and cutting-edge technologies like artificial intelligence (AI), data lakes, analytics, and the Internet of Things (IoT), Amazon hopes to assist businesses and governmental organizations in cutting expenses while improving their flexibility and competitiveness.

Amazon is well-known for a wide range of products, some of which are highly well-liked, in addition to its primary offerings, which include an e-commerce platform and more than 200 cloud services. These consist of Alexa (smart speakers), Fire (tablets and TVs), Kindle (e-book readers), and Twitch (live streaming services). The business is well known for its innovative capabilities and strong focus on the needs of its customers. (techtarget.com , What is Amazon? Definition and company history of Amazon.com)

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Task 1

Throughout the years, Amazon has proven themselves as one of the giants in terms of prioritizing customers' satisfaction through the variety of services they offer. One of these services is managing their customers' information such as their interests in the product, previous purchases, suggesting products based on previous purchases and interests, and also managing their operations such as tracking their supply chain. This can be done by implementing information systems. According to (britannica.com, 17 jul, 2025) introducing information system,

> an integrated set of components for collecting, storing, and processing data and for providing information, knowledge, and digital products. Business firms and other organizations rely on information systems to carry out and manage their operations, interact with their customers and suppliers, and compete in the marketplace. Information systems are used to run interorganizational supply chains and electronic markets.

Amazon has developed their own system. Managing their operations and interaction with customers efficiently. According to a study on Amazon conducted by Md Abdullah Al Imrain (researchgate.net , april,2014), "Amazon Web Service (AWS) and Simple Storage Service (SS) are the main system developed by amazon. Through this system amazon can maintain its vast number of products and millions of active customers."

Amazon equipped themselves with a wide range of robust engines alongside their own cloud platform such as their recommendation engine. Amazon's recommendation system leverages item attributes, consumer activity, and preferences to provide personalized recommendations through data analysis and cutting-edge technologies. (baeldung.com , march 18, 2024)

In terms of Amazon web service, it has helped Amazon and other businesses in various aspects of their operations by removing the need for pricey on-premises data centers and equipment, AWS enabled companies to access scalable and affordable cloud infrastructure and services. Amazon's customer-focused strategy was essential to AWS's success. They paid attention to business problems and used AWS's pay-as-you-go and flexible model to solve them. Businesses of all sizes were able to scale their operations with ease by this customer-focused design, which also decreased IT expenses and increased productivity. (institutefordigitaltransformation.org , Oct 19, 2023).

Additionally, Amazon was able to replace its traditional tape backup systems to move to cloud-based Amazon S3 storage. By replacing their backup tape infrastructure with cloud-based Amazon S3 storage, the company was able to do away with backup software and achieve a 12X speed gain, which in some cases reduced restore time from approximately 15 hours to 2.5 hours. (aws.amazon.com , Amazon.com case study, 2014)

On the other hand, Amazon has invested in various technologies to elevate customers' experience such as recommendation engines. According to brainforge.ai , (how amazon uses big data), "Amazon's recommendation system updates every 10 minutes, processing new customer signals to surface the most relevant products.

"Thirty-five percent of Amazon's sales come from recommendations - that's over $150 billion in annual revenue driven purely by data-powered suggestions"

The Amazon recommendation algorithm analyzes several data layers simultaneously, including items viewed in the same session, products frequently purchased together, past purchase trends, and similar customer preferences across demographics. Amazon improves the customer's shopping experience by using data and machine learning to offer tailored product recommendations. This promotes client loyalty in addition to increasing sales. (brainforge.ai , how amazon uses big data) , (institutefordigitaltransformation.org , Oct 19, 2023)

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Task 2:

Human Computer interaction is “ a field of study that focuses on the interaction between a human (the person using the system) and a computer system (the machine or network of machines that run the system).” (blog.logrocket.com , Aug 7, 2024)

For a company like Amazon, having a user friendly interface is essential to ensure smooth interactions for its customers. Improving the usability of a website can be accomplished by using heuristic evaluation. Which is a process by which usability specialists assess a system using accepted heuristics or usability principles. (linkedin.com , A Heuristic Evaluation of the Amazon Website: Unpacking User Experience, Jan 22, 2024).

The Heuristic evaluation consists of 10 general principles:

1- Users should always receive timely, relevant input from the design to keep them updated on what's happening.

2- Users should be able to understand the design. Instead of using internal jargon, use terms, expressions, and ideas that the user is familiar with. Make information appear in a logical and natural sequence by adhering to real-world conventions.

3- Users frequently make mistakes when doing actions. They require a "emergency exit" that is clearly defined so they can leave the undesirable action without going through an extended process.

4- Users should not have to question whether various terms, circumstances, or behaviors have the same meaning. Observe industry and platform standards.

5- Effective error messages are crucial, but the finest designs take care to avoid issues in the first place. Error-prone conditions should be removed or checked for and users should be given the chance to confirm before proceeding.

6- Reduce the amount of memory used by the user by making options, actions, and elements visible. Information should not have to be remembered when the user navigates through the interface. When needed, information that is necessary to use the design (such as menu items or field labels) should be readily available.

7- In order to accommodate both beginner and experienced users, shortcuts—which are hidden from novice users—may accelerate the interaction for the expert user. Permit users to customize common activities.

8- Information that is rarely used or useless should not be included in interfaces. Each additional unit of information in an interface reduces the relative visibility of the pertinent units of information by competing with them.

9- Error messages should accurately describe the issue, offer a workable solution, and be written in simple terms (no error codes).

10- The system should ideally not require more explanation. To assist users in understanding how to accomplish their jobs, documentation could be required.

(nngroup.com , 10 Usability Heuristics for User Interface Design, April 24, 1994 · Updated Jan. 30, 2024)

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Following the heuristic evaluation principles:

1- With features like real-time order tracking and immediate changes on the shopping cart, Amazon is excellent at giving users clear feedback on their activities. Further improvements can be done on Increasing visibility during times of high traffic to make sure users are aware of any problems or delays.

2- Amazon's terminology is intuitive since it fits with users' mental models. Improvements can be done by consistently improving wording, particularly in notifications and product descriptions, to minimize any possible misunderstandings.

3- Amazon makes it simple for customers to go back and forth, reverse actions, and edit orders before completing them. For further improvements, users could accidentally remove things from their shopping cart, amazon should consider making the "undo" button more noticeable.

4- Amazon keeps its layout and design components same on both its mobile and online platforms. For more improvements , Amazon should consider updating design components often to conform to changing industry standards while making sure users have a smooth transition.

5- Amazon successfully employs error prevention strategies including address validation and password strength indicators. For more improvements, Amazon should add more context specific error prevention warnings to the checkout process to help users and reduce the likelihood of errors.

6- Amazon makes it easier for customers to remember information by providing straightforward product details. However, To reduce the cognitive pressure during the decision making process,Amazon should make critical information like shipping costs more visible.

7- With features like one click shopping and sophisticated search capabilities, Amazon serves both inexperienced and experienced consumers. However, for more improvements, Amazon should increase the visibility of sophisticated features for consumers who might not be aware of time saving possibilities.

8- Amazon has a simple, minimalistic style that puts an emphasis on utility and information. However, To improve the overall user experience, particularly for users with weaker internet connections, Amazon should continuously optimize site loading speeds.

9- During the checkout process, Amazon offers users clear error notices and instructions on how to fix problems.For more improvements, Amazon should Incorporate extra user friendly tools, like chat assistance, to help users instantly in error-prone circumstances.

10- Amazon provides customer service alternatives and a comprehensive help center. However, for more improvements, Improve the website's promotion of self-help tools by encouraging visitors to look for solutions on their own before contacting assistance.

(linkedin.com , A Heuristic Evaluation of the Amazon Website: Unpacking User Experience, Jan 22, 2024).

Task 3

Organisations in general are required to comply with social, legal, and ethical considerations. According to londonsba.org.uk (the importance of ethics in business), " a company with strong ethical principles will naturally earn the trust of its customers, employees, and investors. This trust is the cornerstone of reputation, which is invaluable in the business world."

In terms of amazon, amazon ensures the safety of their customers data. Maintaining their reputation among top e-commerce platforms across the world. According to aws.amazon.com talking about GDPR compliance, "our highest priority is securing customer data, and we implement rigorous technical and organizational measures to protect its confidentiality, integrity, and availability, regardless of which AWS Region the customer has selected."

By adhering to the compliance, amazon maintained their customers trust and loyalty. Amazon also offers significant features for their customers such as the ability to control their personal data:

> Customers have control of their customer data. With AWS, customers can determine where their customer data will be stored, including the type of storage and geographic region of that storage. Choose the secured state of their customer data, and Manage access to their customer data and AWS services and resources through users, groups, permissions and credentials that customers control. (aws.amazon.com , GDPR compliance)

Amazon and many other companies can benefit from complying with social, legal, and ethical considerations. GDPR compliance lowers the risk of data breaches, improves data security, and increases customer trust. It improves operational efficiency, simplifies data management, and strengthens brand reputation. A competitive edge is also produced by global compliance with privacy regulations. (atlan.com Benefits of GDPR compliance, Oct 23, 2024)

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Task 4:

Amazon is known for its vision of adopting and implementing technologies to enhance their operations. For the last 3 years, amazon has adopted various technology solutions to elevate the customers experience and increase the productivity of its operations such as their warehouse operations.

In 2023, Amazon integrated a feature called (Quoting Feature). Customers can create unique quotes for Amazon Business bulk orders using the new Integrated Quoting tool. For large orders placed on Amazon Business, Integrated Quoting offers an automatic quote request. When [the customer] is buying more than 999 things or more than $10,000 in total, this function is perfect. [the customer] easily places an electronic purchase order after reviewing the Amazon Business bid. To give [the customers] additional choices, Fairmarkit incorporates this Integrated Quoting capability with Coupa and other e-procurement platforms. Additionally, Amazon has also integrated 3 way match feature in the amazon business mobile application. 3 Way Match enables clients to speed up the purchasing process by just scanning the barcode on a box, which includes receiving products, reconciling orders, and paying invoices. (business.amazon.com , Reshape Recap, Tech Premiere: What's driving change at Amazon Business? AI and beyond)

In logistic operations, Amazon debuted its most innovative and interesting fulfillment center. This delivery station in Shreveport, Louisiana, uses a variety of Amazon's newest robots, such as Proteus, the first mobile robot that is completely autonomous, to help staff members send items more quickly.

This cutting edge facility reflects Amazon's ambition for collaborative technology that helps workers, enhances sustainability and safety, and speeds up delivery. Shreveport is a turning point in the history of logistics, setting the stage for the transformation of logistics through AI and robotics.

Moreover, Amazon launched Vision Assisted Package Retrieval (VAPR), an AI powered system that recognizes the appropriate products for drivers at every stop in the beginning of 2025. With this AI powered system, drivers no longer have to physically arrange packages in the back of their vehicles since VAPR projected green and red lights onto them at each stop.

According to initial studies, drivers reported a 67% decrease in perceived mental and physical exertion, and each journey saved over 30 minutes. (aboutamazon.com , Amazon announces 7 new innovations that power your shopping and delivery experience).

In distribution and delivery, Amazon has announced the implementation of a generative AI mapping called (Wellspring). Wellspring is aiming to increase customer delivery accuracy. With the help of generative AI, this innovative system generates a full delivery solution for millions of locations by utilizing data from dozens of sources, such as satellite imagery, road networks, building footprints, client instructions, information from previous deliveries, and street images. In order to deliver items to consumers where they want them, this technology helps vehicles better navigate a variety of challenging and varied surroundings, such as apartment complexes with multiple buildings or brand new areas that are not yet visible on navigation apps. (aboutamazon.com , Amazon announces 3 AI-powered innovations to get packages to customers faster).

Amazon's vision in implementing emerging technologies is to enhance their operations and achieve customer satisfaction. According to capital-riesgo.es talking about amazon's strategic acquisition, "Amazon has long been investing in the robotics and artificial intelligence sector to streamline its operations, particularly in distribution and logistics centers. enabling robots to perform complex tasks with greater precision and adaptability."

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Task 5:

Implementing decision support systems is essential for organizations to support the decision making process. In terms of Amazon, it utilizes its Recommendation Engine as a Decision Support System to enhance its decision making processes. According to baeldung.com, March 18, 2024 “Amazon’s recommendation system uses advanced technologies and data analysis to leverage customer behavior, preferences, and item characteristics to deliver tailored suggestions.”

The recommendation engine supports Amazon's decision making processes, which provide data driven and actionable insights that go far beyond user recommendations. By analyzing user behavior, it enables the company to accurately forecast demand, optimize inventory control to prevent stockouts and overstocking, and improve marketing campaigns through A/B testing of the recommendation algorithms. (baeldung.com , March 18, 2024)

The recommendation engine also enhances the overall user experience by streamlining decision making for shoppers, leading to greater loyalty and retention. Personalized suggestions help customers discover relevant products faster, reducing search times and the number of clicks required to complete tasks. (researchgate.net , Using Amazon as a case, a mixed-method study to explore the impact of Personalised Recommendation systems on User Experience and Decision-making, dec, 2023).